

EKY PRATAMA

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 15+ years of experience designing, architecting, and scaling web platforms across the full lifecycle, from on-premise infrastructure to modern cloud-native systems. Spent over a decade taking ownership of an existing marketing automation platform and driving its sustained modernisation: migrating it from on-premise infrastructure to AWS, evolving its database architecture through multiple generations, and upgrading its core runtime and UI framework. Took full end-to-end ownership of new business-critical modules from concept to production, and adopted AI-assisted development practices to improve engineering throughput. Known for taking deep, long-term ownership of complex systems and driving them through sustained technical evolution.

SELECTED TECHNICAL HIGHLIGHTS

- Engineered core platform capability for a SaaS product serving clients across government, financial services, and other sectors, with individual clients ranging from hundreds of thousands to millions of mailgroup subscribers.
- Spearheaded the team's adoption of AI-assisted development, upskilling team members and increasing overall engineering throughput.
- Defined and enforced platform performance standards (sub-150ms transaction times, target-excellent Apdex scores, near-zero error rates) with automated New Relic and PagerDuty escalation on sustained threshold breaches.
- Re-architected a high-throughput marketing automation engine from a SQL/in-memory bottleneck to AWS MemoryDB (Redis Streams) + SQS + Lambda, materially improving page load times and downstream message delivery.
- Designed and built EventDesks, a registration platform handling 10,000+ concurrent event registrants. It replaced a legacy module that could not scale past that ceiling.
- Owned end-to-end delivery of a regulatory compliance integration (ACMA Assist API) for SMS Sender ID verification, built through a full AI agent orchestration workflow with Test-Driven Development (TDD): AI-generated specs, multi-agent audit consortium, AI-assisted implementation and testing, with enforced code quality thresholds (Cyclomatic Complexity <10, Cognitive Complexity <15).
- Took a marketing automation platform through 13 years of continuous modernisation: a full data centre → AWS migration, 4 successive database platform migrations (AWS RDS MySQL → Aurora → Aurora Serverless → Aurora Serverless v2), and 2 major PHP runtime upgrades.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

October 2013 – Present

Swift Digital - Marketing Automation Platform | Sydney, Australia

Swift Digital is a SaaS platform serving clients across government, financial services, and other sectors, providing marketing automation across the full event lifecycle (pre-event, during-event, post-event) through a suite of integrated modules. Individual clients range from hundreds of thousands to millions of mailgroup subscribers.

AI-Assisted Development Adoption - Team Leadership

- Spearheaded the team's adoption of AI-assisted development, upskilling team members on correct, effective use of these tools to help the team adopt them safely and productively, increasing overall engineering throughput.

Profile 360 - Unified Engagement Tracking Platform

- Designed and built the platform's unified engagement model (full-stack, PHP/jQuery), consolidating email, SMS, event, and survey activity into a single customer profile that became the foundation for marketing automation and behavioural triggers.
- Tracked engagement at a granular level across channels: email (sent, opened, link-level clicks), SMS (sent, replies), event registration/rejection, and survey participation, establishing the platform's source-of-truth dataset for end-user engagement.
- Designed and built an embeddable JavaScript tracking snippet ("Goal Reached") deployable on any third-party website, detecting URL-based conversions and securely reporting them via encrypted user identifiers to a Swift Digital API.

Trigger & Campaign Builder - SaaS Marketing Automation Engine

- Took full ownership as main developer of a drag-and-drop campaign orchestration interface (built on JSPlumb, then novel for this use case) paired with a rules-based Trigger module for time-based (e.g. no response after N days) and event-based (e.g. link clicks, opens, subscribe/unsubscribe) automated business logic, plus a custom daemon to process both trigger types in real time.
- Re-architected the processing pipeline to resolve a high-throughput bottleneck: replaced an initial SQL + in-memory table approach with AWS MemoryDB (Redis Streams), chaining events through Amazon SQS to AWS Lambda to execute downstream actions (e.g. individual email/SMS delivery jobs per the organiser's orchestration). This improved page load times and user experience within SQS payload limitations.

Bulk SMS Delivery Redesign

- Eliminated a critical bulk SMS delivery bottleneck: redesigned sequential, one-by-one sending into asynchronous, threshold-based batch calls (e.g. 10,000 per call) against the existing publication daemon's API, with job completion tracked via sent-count/total-count reconciliation. This enabled timely delivery of urgent and emergency notifications for jobs ranging from a single SMS to millions.

Composer & Landing Pages - WYSIWYG Drag-and-Drop Builder

- Active contributor on the platform's Composer module, a WYSIWYG drag-and-drop email builder with a widget-based block canvas used to design and send email campaigns.
- Extended the Composer canvas engine to launch Landing Pages, a new module for web call-to-action forms, reusing the existing canvas and a subset of widgets while building new widgets specific to landing page use cases. Served as main developer for the initial release, then continued as a contributor alongside other engineers as the module matured.

EventDesks - Scalable Event Management Platform

- Designed and developed end-to-end as successor to the legacy Events Classic module, resolving a hard scaling ceiling at 10,000+ registrants; built to handle thousands of registration requests per minute for onsite and online events, with capacity-aware registration logic and organiser-configurable waitlisting to prevent overbooking under concurrent load.
- Designed an extensible payment gateway integration layer (eWay, PayWay, SecurePay, Stripe, and others) and integrated leading video conferencing platforms (GoToWebinar, Zoom, Microsoft Teams) for online events.
- Delivered custom reporting, configurable registration form fields, and one-click event cloning, with fast report generation at 10,000+ registrant scale, a use case that previously bottlenecked Events Classic.
- Built a WYSIWYG drag-and-drop registration form builder for EventDesks, enhancing the same builder engine used for Composer and Landing Pages while developing a new rendering layer specific to event registration forms.

Survey Module - Accessibility Compliance & Modernisation

- Took full ownership of bringing the platform's legacy survey module into WCAG accessibility compliance (WCAG 2.0, later 2.2) for government clients, remediating auditor-identified issues across both frontend markup and backend form logic.
- Leading a ground-up rewrite of the survey module on SurveyJS (in progress), architected for 10,000+ respondents per survey, with AWS MemoryDB-backed save-and-resume and an async S3 upload pipeline for respondent file submissions.

ACMA Assist Integration - Regulatory Compliance & AI Agent Orchestration

- Owned end-to-end design and delivery of an integration with the ACMA Assist API, enabling organisations to register and verify SMS Sender IDs in compliance with new Australian regulatory requirements mandating verified business registration for branded SMS sending.
- Built an automated registration and internal approval workflow, including conflict detection that flags staff when a requested Sender ID is already registered elsewhere, and orchestrated the asynchronous approval lifecycle pending ACMA's external sign-off.
- Delivered the full build through an end-to-end AI agent orchestration workflow using OpenCode and GitHub Copilot (VS Code) with Test-Driven Development (TDD): AI-generated specifications, audited by a multi-agent AI consortium for independent code review, followed by AI-assisted implementation, TDD-driven unit test generation, and browser/regression testing via Playwright MCP.
- Enforced concrete code quality gates throughout (Cyclomatic Complexity under 10, Cognitive Complexity under 15) and adherence to OWASP secure coding standards across the AI-generated and human-reviewed codebase.

Platform Security Engineering & Compliance

- Drove the platform's ongoing OWASP compliance and security posture as part of a defence-in-depth strategy combining AWS WAF (first layer, including geo-restriction and CAPTCHA challenge rules) with Trend Micro Deep Security IDS/IPS (second layer), and contributed to remediation following regular third-party penetration tests.

- Led and defined the platform's performance standards (sub-150ms transaction response times, target-excellent Apex scores, near-zero error rates) and built the monitoring and alerting around them: automated New Relic and PagerDuty escalation triggered after 5 consecutive minutes of threshold breach, feeding into a formal on-call rotation covering security events, platform slowness/downtime, and critical production bugs.
- Contributed to the platform's ISO 27001 certification as part of the compliance team, executing assigned controls and remediation tasks.
- Hold a designated access control role for Trend Micro Deep Security and New Relic, approving third-party plugin adoption requests from other engineers and managing user provisioning/deprovisioning on both platforms. This role is subject to independent audit every 3 months.

Infrastructure & Platform Modernisation

- Contributed to a company-wide migration from on-premise data centre infrastructure to AWS (2016); re-engineered image storage and delivery, migrating from database-stored BLOBs to direct AWS S3 object storage for scalable, public image delivery across email and web features.
- Later established a CDN using Amazon CloudFront, built on top of the S3 storage foundation, extending fast, scalable content delivery beyond images to broader platform assets, with geo-restriction blocking high-risk countries via CloudFront and AWS WAF.
- Contributed as part of the team driving successive production database migrations across multiple generations: on-premise MySQL to AWS RDS MySQL to Aurora to Aurora Serverless to Aurora Serverless v2; led the database performance engineering (query rewrites, indexing, schema design) using New Relic for data-driven diagnosis of version-related regressions.
- Main contributor on sequential PHP runtime migrations (PHP 5→7, then 7→8), resolving deprecated functions, type-system changes, and breaking changes; used GitHub Copilot to accelerate PHP 8 remediation and introduced AI-assisted unit testing/TDD practices now standard on the team.
- Served as main developer driving a platform-wide UI migration from custom HTML/CSS to Bootstrap, adopted early when the framework was new, implementing front-to-back UI changes based on design guidelines and Bootstrap documentation.

SKILLS

- **Agentic AI Architecture Engineering:** Multi-agent orchestration, AI-generated specification design, OpenCode, GitHub Copilot, Hermes Agents, MCP, TDD, Automated Testing
- **Cloud Infrastructure:** AWS (S3, CloudFront, RDS, Aurora Serverless, MemoryDB/Redis Streams, SQS, Lambda, CloudFormation)
- **Business Process Optimisation:** BPMN, visual workflow orchestration
- **Database Engineering:** MySQL, query optimisation, schema design
- **Security & Compliance:** OWASP, ISO 27001, AWS WAF, Trend Micro IDS/IPS, Penetration Test Remediation, WCAG 2.0/2.2
- **Operations & Methodology:** PagerDuty, New Relic, Agile/Scrum SDLC, Incident Response & On-Call, Git
- **Languages & Frameworks:** PHP, JavaScript, jQuery, CSS/Sass, Bootstrap, SurveyJS

EDUCATION

Master of Computer Science (Software Engineering)

University of Wollongong

Bachelor of Computer Science

University of Wollongong

CERTIFICATIONS

Common Cyber Security Threats and Mitigation Strategies

TAFE | June 2022 | Certificate ID: Q6fqdNWMve

AWS Technical Essentials

Amazon Web Services | June 2017 | <https://goo.gl/d25WSR>

References available upon request